Dr. D. Y. Patil Educational Federation's

Dr. D. Y. Patil College of Engineering and Innovation

# DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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## PERSONAL FINANCE ADVISOR BOT: AI-POWERED PERSONAL FINANCE MANAGEMENT PLATFORM

**Project Description :-**

**Personal Finance Advisor Bot is an AI-powered personal finance management and advisory platform designed to help users manage their income, expenses, budgets and savings goals in one place.**

**The platform provides a personalized financial workspace where users can securely create an account, record their income and expenses, organize transactions into categories, create monthly budgets and monitor savings goals. The dashboard provides a clear overview of the user's financial activity through summaries, charts and financial indicators.**

**The system also provides an intelligent AI Financial Advisor that uses the user's financial records as context to provide personalized educational financial guidance and recommendations. The AI service supports Google Gemini and OpenAI, while a rule-based fallback provides recommendations when an external AI**

**service is unavailable or not configured.**

**The application includes financial health analysis, category-wise spending analysis, budget monitoring, savings progress, AI advisor chat and AI recommendation history. Users can also generate monthly financial reports and export them in CSV and PDF formats.**

**The project is developed as a production-oriented web application using Python, Flask, Jinja2, SQLAlchemy, Flask-Login, Flask-WTF, Bootstrap 5, JavaScript and Chart.js, with SQLite for local development and support for PostgreSQL-compatible deployment.**

**Technology & AI :-**

**Frontend: HTML5 / CSS3 / Jinja2 Templates / Bootstrap 5 / JavaScript / Chart.js / Responsive Financial Dashboard**

**Backend: Python / Flask / Flask Application Factory / Flask Blueprints / SQLAlchemy ORM**

**Database: SQLite for local development / SQLAlchemy ORM / PostgreSQL-compatible database support**

**Authentication: Flask-Login / Werkzeug Password Hashing / Secure Sessions / Protected Routes**

**Security: Flask-WTF CSRF Protection / Password Hashing / Environment Variables for API Keys / User Ownership Isolation**

**AI: Google Gemini API / OpenAI API / Rule-Based AI Fallback / AI Recommendation History**

**Visualization: Chart.js / Financial Dashboard Charts / Category-wise Spending / Budget Status Indicators / Savings Progress**

**Testing: Pytest / Automated Application Tests**

**Deployment: Docker / Gunicorn / Render / Health Check Endpoint**

**Version Control: Git / GitHub**

**Key Features :-**

**• Secure user registration and login system**

**• Personal finance dashboard**

**• Income management and tracking**

- **Expense management and tracking**
- **Category-based financial organization**
- **Custom expense categories**
- **Monthly budget planning**
- **Automatic budget generation based on user profile and financial information**
- **Savings goal management and progress tracking**
- **AI-powered financial advisor**
- **AI advisor chat**
- **Personalized financial recommendations based on user's financial records**
- **Google Gemini AI integration**
- **OpenAI integration**
- **Rule-based AI fallback when external AI services are unavailable**
- **Financial health analysis and spending insights**
- **Dashboard financial charts using Chart.js**
- **Monthly financial reports**
- **CSV and PDF report export**
- **AI recommendation history**
- **Secure user ownership and protected financial records**
- **Automated application testing**
- **Deployment health-check endpoint**

**Scenarios :-**

**Scenario 1: Student Managing Monthly Expenses**

**A student registers on the Personal Finance Advisor Bot and enters their monthly income from allowance,**

scholarship or part-time work. The student records expenses such as food, transportation, education and entertainment. The dashboard summarizes the financial information and displays category-wise spending. The AI advisor provides educational suggestions for controlling unnecessary expenses and improving savings.

Scenario 2: Salaried User Planning a Monthly Budget

A salaried user enters their monthly salary and regular expenses. The user creates budgets for categories such as food, transportation, bills, shopping and entertainment. The system compares actual spending with the planned budget and displays budget status indicators. The AI advisor uses the user's financial records to provide personalized educational recommendations for better money management.

Scenario 3: Freelancer Managing Variable Income

A freelancer records income and expenses that may vary from month to month. The system analyzes the available financial information and provides a suitable budget structure for essential expenses, lifestyle spending and savings. The user can monitor spending patterns and savings progress through the dashboard. The AI advisor provides general financial guidance based on the user's recorded information.

Scenario 4: User Creating a Savings Goal

A user wants to save money for a specific financial goal. The user creates a savings goal and enters the target amount, current savings and optional deadline. The system tracks the progress toward the target and displays the savings status. The AI advisor can provide educational suggestions related to budgeting and achieving the savings goal.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Define the objectives of the Personal Finance Advisor Bot, identify target users and finalize the major modules such as authentication, income, expenses, categories, budgets, savings goals, dashboard, AI advisor and reports.

Milestone 2 – Environment Setup & Backend Development

Set up the Python and Flask development environment. Implement the Flask application factory, Blueprint-based application structure, SQLAlchemy database integration and required financial data models.

**Milestone 3 – Authentication & Security Layer**

**Develop user registration and login functionality. Implement password hashing, protected routes, session management, CSRF protection, environment-based configuration and user ownership isolation.**

**Milestone 4 – Finance Management Modules**

**Implement the core financial modules including income management, expense management, categories, monthly budgets and savings goals. Develop CRUD operations so users can create, view, update and manage their personal financial records.**

**Milestone 5 – AI Advisor, Dashboard & Reports**

**Integrate Google Gemini and OpenAI as supported AI providers along with a rule-based fallback. Implement AI advisor chat, recommendation history, financial dashboard charts, budget generation, financial health analysis, monthly reports and CSV/PDF export.**

**Milestone 6 – Testing, Optimization & Deployment**

**Perform automated testing using Pytest. Test registration, login, protected routes, income and expense operations, user ownership, AI fallback, budget generation, dashboard calculations, reports and database persistence. Finally, deploy the application using a production web-server setup with a health-check endpoint.**

**System Workflow :-**

**User Registration / Login**

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**Personal Finance Dashboard**

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**Add Income & Expenses**

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**Organize Transactions into Categories**

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**Create / Manage Monthly Budgets**

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**Create & Track Savings Goals**

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**Financial Data Analysis**

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**AI Financial Advisor**

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**Google Gemini / OpenAI / Rule-Based Fallback**

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**Personalized Educational Financial Guidance**

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**Generate Monthly Financial Report**

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**View / Export PDF / CSV**

**Conclusion :-**

**Personal Finance Advisor Bot provides an intelligent and interactive platform for personal finance management by combining traditional financial tracking with Artificial Intelligence.**

**The application allows users to record income and expenses, organize transactions, create monthly budgets, track savings goals and analyze their financial activity through an interactive dashboard. The AI Financial Advisor uses the user's financial records to provide personalized educational guidance and recommendations.**

**The system also includes a rule-based fallback mechanism so that basic financial recommendations can remain available when external AI services are unavailable. With its Flask-based backend, SQLAlchemy**

**database layer, secure authentication, Chart.js visualizations, AI integration, automated testing and deployment support, the project demonstrates the practical application of modern web technologies and Artificial Intelligence in the FinTech domain.**

**Expected Outcome :-**

**The Personal Finance Advisor Bot aims to help users better understand and manage their personal finances through a single integrated platform.**

**The application provides automated income and expense tracking, budget planning, savings goal management, financial visualization, financial health analysis, AI-powered educational recommendations and monthly financial reports.**

**The project demonstrates practical expertise in:**

- **Artificial Intelligence and Generative AI**
- **Python and Flask**
- **SQLAlchemy and Database Management**
- **Web Application Development**
- **User Authentication and Security**
- **Google Gemini and OpenAI API Integration**
- **Rule-Based AI Systems**
- **Financial Data Visualization**
- **Automated Testing using Pytest**
- **Git and GitHub**
- **Cloud/Web Deployment**